NAME: _____ Geometry (FL B.E.S.T.)

12.2 Independent and Dependent Events

Two events are **independent** if the outcome of one has no effect on the probability of the other (a coin flip doesn't change a die roll) — then $P(A \text{ and } B) = P(A) \cdot P(B)$, using the *same* probability for B regardless of A. Two events are **dependent** if one outcome changes the probability of the next — the classic signal is drawing **without replacement**, where removing an item changes the totals for the next draw. This worksheet practises multiplying along both types of chains, recognizing which type a situation is, and extending to three or more sequential events.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A coin is flipped and a die is rolled. Find $P(\text{tails and even})$.

2. Bag: 5 red, 5 blue. With replacement, find $P(\text{both red})$.

3. Same bag. Without replacement, find $P(\text{both red})$.

4. Two rolls of a die: independent or dependent?

5. Without replacement, find $P(\text{two hearts})$.

6. Two dice rolled. Find $P(\text{both even})$.

7. 2 red, 8 blue. Without replacement, find $P(\text{both red})$.

8. 2 red, 8 blue. With replacement, find $P(\text{both red})$.

9. Three fair coins. Find $P(HHH)$.

10. Why is a tree diagram useful for sequential (multi-stage) events?

11. Without replacement, find $P(\text{ace then king})$.

12. The phrase "without replacement" usually signals which type of events?

13. Independent events: $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

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14. Three marbles are drawn without replacement from a bag of 3 red and 5 blue. Find $P(\text{all three red})$.

15. A bag has 6 white and 4 black balls. Three balls are drawn without replacement. Find $P(\text{exactly the order white, white, black})$, as a fraction in lowest terms.

✓ Answer Key

1. $\frac{1}{4}$
2. $\frac{1}{4}$
3. $\frac{2}{9}$
4. Independent
5. $\frac{1}{17}$
6. $\frac{1}{4}$
7. $\frac{1}{45}$
8. $\frac{1}{25}$
9. $\frac{1}{8}$
10. It organizes every branching outcome and its probability, making them easy to multiply and add
11. $\frac{4}{663}$
12. Dependent
13. $\frac{1}{6}$
14. $\frac{1}{56}$
15. $\frac{1}{6}$